

AP CALCULUS AB – UNIT 2

Differentiation: Definition and Fundamental Properties

Expanded Complete-Coverage Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Compute and interpret average and instantaneous rates of change.
- Use the limit definition of the derivative and write tangent-line equations.
- Estimate derivatives and analyze differentiability and continuity from multiple representations.
- Differentiate algebraic, exponential, logarithmic, and trigonometric functions using foundational, product, and quotient rules.
- Review every idea, theorem hypothesis, representation, procedure, and common misconception appearing in the website lessons.

Complete Lesson Coverage Audit

Every website lesson is given its own mastery section below. Each section contains the complete lesson objectives, strategic reminders, a solved model, and foundation-to-challenge practice. The final synthesis sections mix lessons so that students must select procedures rather than follow a stated template.

Complete Website Lesson Coverage Map

Lesson	Website title	Ideas explicitly reviewed
2.1	Rates of Change and Tangent Lines	Calculate average rate of change from formulas, tables, graphs, and contexts.; Interpret secant-line slope and tangent-line slope with correct units.; Use shrinking intervals to motivate and estimate instantaneous rate of change.
2.2	Tangent Lines and the Derivative	Represent a derivative at a point and a derivative function as limits of difference quotients.; Translate among $f'(a)$, y' , $\frac{dy}{dx}$, and related notation.; Use a derivative value to write the equation of a tangent line.
2.3	The Derivative: Graphs, Differentiability, and Continuity	Estimate a derivative from a graph, numerical table, or local data.; Compare forward, backward, and centered differences.; Judge the reliability of an estimate using interval size and local smoothness.; Explain why differentiability at a point implies continuity there.; Identify discontinuities, corners, cusps, and vertical tangents as sources of nondifferentiability.; Determine parameter values that make a piecewise function continuous and differentiable.

2.4	Basic Derivative Rules	Differentiate powers x^n for integer, rational, and permitted real exponents.; Rewrite radicals and reciprocals as powers before differentiating.; Evaluate derivatives and tangent slopes without unnecessary expansion.; Apply constant, sum, difference, and constant-multiple rules efficiently.; Differentiate polynomial and power combinations term by term.; Interpret linearity of differentiation and avoid distributing rules that do not exist.; Differentiate the four foundational transcendental functions in radians.; Combine these derivatives using linearity and power rules.; Use exact values, domains, and context when evaluating derivatives.; Differentiate $\tan x$, $\cot x$, $\sec x$, and $\csc x$.; Evaluate derivatives at exact angles and identify domain restrictions.; Combine the remaining trigonometric derivative rules with prior linearity, Product, and Quotient Rules.
2.5	Product and Quotient Rules	Differentiate products of differentiable functions.; Use values and derivative values from tables or graphs in the Product Rule.; Choose between expanding first and applying the Product Rule.; Differentiate quotients of differentiable functions.; Apply the Quotient Rule to symbolic, tabular, and contextual representations.; Respect denominator restrictions and simplify strategically.

Unit Knowledge, Definitions, and Theorem Library

Key Knowledge, Formulas, and Theorems

- Average and instantaneous rates of change are secant and tangent slopes, with units of output per input.
- Point definition: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$; equivalent form: $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.
- Derivative notation includes $f'(x)$, y' , $\frac{dy}{dx}$, $D_x f$, and higher derivatives such as $f''(x)$.
- Tangent line: $y - f(a) = f'(a)(x - a)$. The normal slope is the negative reciprocal when the tangent slope is finite and nonzero.
- Derivative estimates from tables should use a centered difference when suitable data occur on both sides.
- Differentiability implies continuity. Corners, cusps, vertical tangents, and discontinuities are standard sources of nondifferentiability.
- The derivative function records the slope of the original graph; zeros correspond to horizontal tangents but do not automatically classify extrema.
- Foundational rules include constant, power, sum, difference, and constant-multiple rules together with derivatives of exponential, logarithmic, and trigonometric functions.
- Product and quotient rules preserve function order: $(fg)' = f'g + fg'$ and $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
- Values of products, quotients, and derivatives can be computed from formulas, graphs, or tables without knowing explicit expressions.

Strategy and Precision Checklist

1. Name the representation and the requested quantity before calculating.
2. State all hypotheses before using a theorem.
3. Preserve exact values in non-calculator work and use radians for trigonometric calculus.
4. A graph or table supports an estimate; an analytic proof requires a definition, theorem, or valid algebraic argument.
5. Contextual conclusions must include the instant, sign, units, and meaning.
6. Candidate conditions such as a zero derivative, zero second derivative, or zero denominator require sign/domain verification before a conclusion.

Lesson 2.1: Rates of Change and Tangent Lines**Complete objectives**

- Calculate average rate of change from formulas, tables, graphs, and contexts.
- Interpret secant-line slope and tangent-line slope with correct units.
- Use shrinking intervals to motivate and estimate instantaneous rate of change.

Strategic reminders

1. An average rate describes net change across an interval. It does not claim that the rate stayed constant inside the interval.
2. To approximate a rate at $x = a$, use secant slopes over intervals with endpoints increasingly close to a . A symmetric interval often reduces one-sided bias.
3. In contextual answers, state the quantity, sign, units, and time or input value. A negative rate means the dependent variable is decreasing with respect to the independent variable.

Solved Example: Secant slope from a formula

For $s(t) = t^3 - 4t$ meters, find the average velocity on $[1, 3]$.

Answer and Reasoning

$$s(3) = 15 \text{ and } s(1) = -3, \text{ so } \frac{15 - (-3)}{3 - 1} = 9 \text{ meters per second.}$$

Lesson Mastery Practice

1. Find the average rate of change of $f(x) = 3x^2 - 2$ on $[1, 4]$.

Foundation

2. For $s(t) = t^3 - 6t^2 + 9t$, find the average velocity on $[1, 2]$ and on $[1, 1.5]$.

Core

3. For $f(x) = x^2 - 5x$, determine all intervals $[a, a + 2]$ on which the average rate is 3.

Advanced

4. For $f(x) = |x|$, compute the average rates on $[-h, 0]$ and $[0, h]$ for $h > 0$. What do they predict at 0? **Challenge**

AP-Style Reasoning Check

The average rate of change of f on $[a, b]$ is

(A) $\frac{f(a)-f(b)}{b-a}$

(B) $\frac{f(b)-f(a)}{b-a}$

(C) $f(b) - f(a)$

(D) $\frac{b-a}{f(b)-f(a)}$

Lesson 2.2: Tangent Lines and the Derivative**Complete objectives**

- Represent a derivative at a point and a derivative function as limits of difference quotients.
- Translate among $f'(a)$, y' , $\frac{dy}{dx}$, and related notation.
- Use a derivative value to write the equation of a tangent line.

Strategic reminders

1. The base point must remain fixed in a derivative-at-a-point limit. In the h -form, the nearby input is $a + h$; in the $x \rightarrow a$ form, the denominator is $x - a$.
2. Notation communicates both the function and the variable of differentiation. For example, $\frac{dP}{dt}$ is the rate of change of P with respect to t .
3. A derivative value supplies only the tangent slope. To write the tangent line, also use the point $(a, f(a))$.

Solved Example: Definition at a point

Use the limit definition to find $f'(3)$ for $f(x) = x^2 - 4x$.

Answer and Reasoning

$$\frac{(3+h)^2 - 4(3+h) - (-3)}{h} = \frac{2h+h^2}{h} = 2 + h, \text{ so } f'(3) = 2.$$

Lesson Mastery Practice

5. Write $f'(5)$ using the h -form of the derivative definition.

Foundation

6. Use the definition to find $f'(1)$ for $f(x) = x^2 + 3x$.

Core

7. For $f(x) = \frac{1}{x}$, use the definition to derive $f'(x)$ for $x \neq 0$.

Advanced

8. If $f'(a)$ exists, prove $\lim_{h \rightarrow 0} \frac{f(a+2h) - f(a)}{h} = 2f'(a)$.

Challenge

AP-Style Reasoning Check

Which limit equals $f'(4)$?

(A) $\lim_{h \rightarrow 0} \frac{f(4+h) - f(h)}{4}$

(C) $\lim_{h \rightarrow 4} \frac{f(h) - f(4)}{h}$

(B) $\lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$

(D) $\lim_{x \rightarrow 0} \frac{f(x) - f(4)}{x-4}$

Lesson 2.3: The Derivative: Graphs, Differentiability, and Continuity

Complete objectives

- Estimate a derivative from a graph, numerical table, or local data.
- Compare forward, backward, and centered differences.
- Judge the reliability of an estimate using interval size and local smoothness.
- Explain why differentiability at a point implies continuity there.
- Identify discontinuities, corners, cusps, and vertical tangents as sources of nondifferentiability.
- Determine parameter values that make a piecewise function continuous and differentiable.

Strategic reminders

1. **Derivative Estimation.** Use points close to the target, but not so close that rounding error dominates. When a table is rounded, several nearby estimates should be compared.
2. **Derivative Estimation.** A centered difference balances information from both sides and cancels some first-order error for smooth functions.
3. **Derivative Estimation.** At a corner or cusp, left and right slope estimates approach different values or become unbounded; reporting one centered number can hide nondifferentiability.
4. **Differentiability and Continuity.** To test a piecewise join, first impose continuity. Only after the function values match should you impose equality of the one-sided derivatives.
5. **Differentiability and Continuity.** A vertical tangent may have a visually well-defined tangent line but its slope is not a finite real number, so the ordinary derivative does not exist.
6. **Differentiability and Continuity.** When justifying, name the exact condition that fails: unequal one-sided slopes, unbounded slope, or lack of continuity.

Solved Example: Derivative Estimation - Centered estimate

Given $f(1.9) = 4.61$ and $f(2.1) = 5.43$, estimate $f'(2)$.

Answer and Reasoning

$$\frac{5.43 - 4.61}{0.2} = 4.1.$$

Lesson Mastery Practice

9. Derivative Estimation: Use $f(2.9) = 8.41$ and $f(3.1) = 9.61$ to estimate $f'(3)$ by a centered difference.

Foundation

10. Derivative Estimation: Using the shared table, estimate $f'(2)$ with $h = 0.01$ and with $h = 0.10$.

Core

11. Derivative Estimation: For a smooth function, forward estimates with $h = .2, .1, .05$ are 3.40, 3.20, 3.10. Predict the derivative and explain the pattern.

Advanced

12. Derivative Estimation: Show algebraically that the centered difference for $f(x) = x^3$ at a equals $3a^2 + h^2$.

Challenge

AP-Style Reasoning Check

Derivative Estimation: Which estimate generally uses data from both sides of a ?

(A) forward difference

(B) backward difference

(C) centered difference

(D) average function value

Lesson 2.4: Basic Derivative Rules**Complete objectives**

- Differentiate powers x^n for integer, rational, and permitted real exponents.
- Rewrite radicals and reciprocals as powers before differentiating.
- Evaluate derivatives and tangent slopes without unnecessary expansion.
- Apply constant, sum, difference, and constant-multiple rules efficiently.
- Differentiate polynomial and power combinations term by term.
- Interpret linearity of differentiation and avoid distributing rules that do not exist.
- Differentiate the four foundational transcendental functions in radians.
- Combine these derivatives using linearity and power rules.
- Use exact values, domains, and context when evaluating derivatives.
- Differentiate $\tan x$, $\cot x$, $\sec x$, and $\csc x$.
- Evaluate derivatives at exact angles and identify domain restrictions.
- Combine the remaining trigonometric derivative rules with prior linearity, Product, and Quotient Rules.

Strategic reminders

1. **Power Rule.** Lower the exponent by one after multiplying by the original exponent. Keep exact fractional exponents unless the problem requests radical form.
2. **Power Rule.** Check where the function is real and defined. For example, $x^{1/2}$ is real for $x \geq 0$, while its derivative $1/(2\sqrt{x})$ requires $x > 0$.
3. **Power Rule.** A limit that matches the derivative definition of a power function can often be evaluated immediately by the Power Rule.
4. **Linearity Rules.** Differentiation is linear: constants can be pulled through the derivative, and sums can be differentiated term by term.
5. **Linearity Rules.** First simplify algebraically when it reduces work, especially when common factors or quotients of powers can be rewritten. Do not invent a rule such as $(fg)' = f'g'$; products require the Product Rule in Topic 2.8.
6. **Linearity Rules.** After differentiating, evaluate only if a particular input is requested. Keeping symbolic structure often reduces arithmetic errors.
7. **Core Transcendental Derivatives.** The standard trigonometric derivative formulas assume radians. Degree measure introduces an additional conversion factor.
8. **Core Transcendental Derivatives.** For sums and constant multiples, differentiate each term separately. Composite functions are postponed to the Chain Rule in Unit 3.
9. **Core Transcendental Derivatives.** Preserve the domain of $\ln x$. A derivative formula evaluated outside the original function domain is meaningless.
10. **Tangent, Cotangent, Secant, and Cosecant.** Keep signs attached to cotangent and cosecant derivative rules. A compact memory pattern is: tangent and secant are positive; cotangent and cosecant are negative.
11. **Tangent, Cotangent, Secant, and Cosecant.** Use unit-circle values to evaluate exactly. Check that the original function is defined at the requested angle before evaluating its derivative.
12. **Tangent, Cotangent, Secant, and Cosecant.** These formulas can be derived from $\tan = \sin/\cos$, $\cot = \cos/\sin$, $\sec = 1/\cos$, and $\csc = 1/\sin$ using earlier rules.

Solved Example: Power Rule - Polynomial power

Differentiate $f(x) = x^7$.

Answer and Reasoning

$$f'(x) = 7x^6.$$

Lesson Mastery Practice

13. Power Rule: Differentiate x^9 .

Foundation

14. Power Rule: Differentiate $f(x) = x^{7/3} - x^{-2}$.

Core

15. Power Rule: Differentiate $f(x) = \frac{x^5+1}{x^3}$ after rewriting.

Advanced

16. Power Rule: Using only the Power Rule, show that the derivative of $x^{m/n}$ is $\frac{m}{n}x^{m/n-1}$ for positive x by differentiating $y^n = x^m$ informally and solving for y' .

Challenge

AP-Style Reasoning Check

Power Rule: $\frac{d}{dx}x^{-5}$ equals

(A) $-5x^{-6}$

(B) $-5x^{-4}$

(C) x^{-6}

(D) $5x^4$

Lesson 2.5: Product and Quotient Rules**Complete objectives**

- Differentiate products of differentiable functions.
- Use values and derivative values from tables or graphs in the Product Rule.
- Choose between expanding first and applying the Product Rule.
- Differentiate quotients of differentiable functions.
- Apply the Quotient Rule to symbolic, tabular, and contextual representations.
- Respect denominator restrictions and simplify strategically.

Strategic reminders

1. **Product Rule.** Name the factors before differentiating. Keep one factor unchanged while differentiating the other, then reverse roles and add.
2. **Product Rule.** If both factors are simple polynomials, expansion can be faster and provides a useful check. If factors have meaningful structure or include transcendental functions, the Product Rule is usually cleaner.
3. **Product Rule.** Table problems require four values at the target: f , f' , g , and g' .
4. **Quotient Rule.** Write the denominator squared first, then construct the ordered numerator. The minus sign is a frequent source of errors.
5. **Quotient Rule.** Before using the rule, check whether algebraic simplification turns the quotient into a sum of powers. Simplifying can be faster, but domain restrictions from canceled factors must still be remembered.
6. **Quotient Rule.** A horizontal tangent occurs where the derivative numerator is zero while the original denominator is nonzero.

Solved Example: Product Rule - Symbolic product

Differentiate $f(x) = x^3 \sin x$.

Answer and Reasoning

$$f'(x) = 3x^2 \sin x + x^3 \cos x.$$

Lesson Mastery Practice

17. Product Rule: Differentiate $x^2 e^x$.

Foundation

18. Product Rule: Differentiate $f(x) = x^4 \ln x$ for $x > 0$.

Core

19. Product Rule: Find $(f^2)'(a)$ in terms of $f(a)$ and $f'(a)$ without invoking the Chain Rule.

Advanced

20. Product Rule: Prove the Product Rule from the limit definition by adding and subtracting $f(x+h)g(x)$.

Challenge

AP-Style Reasoning Check

Product Rule: $(fg)' =$

(A) $f'g'$

(B) $f'g + fg'$

(C) $fg' + g$

(D) $f' + g'$

Unit 2 Difficult and Challenge Synthesis

1. Use the limit definition to derive $\frac{d}{dx}\sqrt{x}$ at a general point $x = a > 0$.

Difficult

2. Prove directly from the derivative definition that differentiability at a implies continuity at a .

Challenge

3. Find a, b so that $p(x) = ax + b$ for $x < 1$ and $p(x) = x^2 - 2x + 4$ for $x \geq 1$ is differentiable at 1.

Hard

4. Let $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, and $g'(2) = 5$. Find the derivative at 2 of $H(x) = \frac{x^2 f(x)}{g(x)}$ without expanding.

Difficult

5. A continuous function has a cusp at $x = 0$, a vertical tangent at $x = 2$, and a removable discontinuity at $x = 4$. At which points can it be differentiable? Explain using one-sided difference quotients.

Hard

6. Differentiate $F(x) = \frac{(x^3 - 1)(e^x + \sin x)}{x^2 + 1}$ and evaluate $F'(0)$.

Challenge

Cumulative AP-Style Checkpoint

1. A quantity $M(t)$ is measured in kilograms and t in hours. Which is a complete interpretation of $M'(5) = -2.4$?
- (A) During the first 5 hours, mass decreases by 2.4 kilograms
(B) At $t = 5$, time decreases at 2.4 hours per kilogram
(C) At $t = 5$ hours, the mass is decreasing at 2.4 kilograms per hour
(D) At $t = 5$, the mass is -2.4 kilograms
2. The limit $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = 7$ directly implies which statement?
- (A) g is linear
(B) $g'(3) = 7$
(C) $g(3) = 7$
(D) $g'(7) = 3$
3. Which statement is always true?
- (A) If f has a vertical tangent at c , then $f'(c)$ is finite
(B) If f is differentiable at c , then f is continuous at c
(C) If f is continuous at c , then f is differentiable at c
(D) If $f'(c) = 0$, then f has a local extremum at c
4. If $F(x) = x^3 \ln x$ for $x > 0$, then $F'(x)$ is
- (A) $3x^2/x$
(B) x^3/x
(C) $3x^2 \ln x + 1/x$
(D) $3x^2 \ln x + x^2$
5. At $x = 1$, $f = 2$, $f' = -3$, $g = 4$, and $g' = 5$. What is $\left(\frac{f}{g}\right)'(1)$?
- (A) $-\frac{11}{8}$
(B) $\frac{1}{2}$
(C) $-\frac{1}{4}$
(D) $\frac{11}{8}$
6. For $f(x) = x^3$, which expression is the average rate of change on $[a, a + h]$?
- (A) $3a^2$
(B) $a^3 + h^3$
(C) $3(a + h)^2$
(D) $3a^2 + 3ah + h^2$
7. If $f(x) = \sqrt{x}$, then $f'(4)$ equals
- (A) 2
(B) 4
(C) $\frac{1}{4}$
(D) $\frac{1}{2}$
8. For $f(x) = |x - 2|$, the left- and right-hand derivatives at 2 are
- (A) -1 and -1
(B) 0 and 0
(C) -1 and 1
(D) 1 and 1
9. Differentiate $f(x) = 5x^7 - 4x^2 - 4$.
- (A) $x(35x^5 - 8)$
(B) $x(5x^5 - 8)$
(C) $35x^6 - 8x + 1$
(D) $x(35x^6 - 8)$
10. Differentiate $(x^2 + 1)(5x - 1)$.
- (A) $5x^2 + 5$
(B) $10x$
(C) $2x(5x - 1)$
(D) $15x^2 - 2x + 5$

11. For $f(x) = 4x^2 - 3x$, find the average rate of change on $[3, 6]$.

- (A) 33 (B) 34
(C) -33 (D) 32

12. If $f(x) = 2x^2 - 1x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$.

- (A) 2 (B) 4
(C) 3 (D) -3

13. A table gives $f(\frac{9}{10}) = \frac{28}{5}$ and $f(\frac{11}{10}) = \frac{32}{5}$. Estimate $f'(1)$ using a symmetric difference.

- (A) 8 (B) 4
(C) $\frac{4}{5}$ (D) 2.0

14. Find $\frac{d}{dx}[\log(x)]$.

- (A) $\log(x) + 1$ (B) $-\frac{1}{x}$
(C) $\frac{1}{x}$ (D) x

15. Differentiate $\frac{x^2 + 1}{x + 2}$.

- (A) $2x$ (B) $\frac{2x}{x+2}$
(C) $\frac{-x^2 + 2x(x+2) - 1}{(x+2)^2}$ (D) $\frac{x^2 + 2x(x+2) + 1}{(x+2)^2}$

16. For $f(x) = 3x^2 + 3x$, find the average rate of change on $[1, 3]$.

- (A) 14 (B) 15
(C) -15 (D) 16

17. Find the tangent line to $y = 3x^2 - 2$ at $x = 2$.

- (A) $y = 12x - 13$ (B) $y = 13x - 16$
(C) $y = 12x - 14$ (D) $y = 12x + 34$

18. A table gives $f(\frac{9}{10}) = \frac{47}{10}$ and $f(\frac{11}{10}) = \frac{53}{10}$. Estimate $f'(1)$ using a symmetric difference.

- (A) 3 (B) $\frac{3}{5}$
(C) 1.5 (D) 6

19. Differentiate $f(x) = 3x^3 + 5x^2 - 3$.

- (A) $x(3x + 10)$ (B) $x(9x + 10)$
(C) $x(9x^2 + 10)$ (D) $9x^2 + 10x + 1$

20. At $x = 4$, $f = 4$, $g = 4$, $f' = 2$, and $g' = -1$. Find $(fg)'(4)$.

- (A) 3 (B) 4
(C) -4 (D) 5

Cumulative Free-Response Synthesis

FRQ1 – Rates, derivative estimates, and tangent model

A differentiable function P models a population in thousands. Selected values are					
t	1.8	1.9	2.0	2.1	2.2
$P(t)$	4.612	4.801	5.000	5.209	5.428

(a) Estimate $P'(2)$ using the best symmetric difference available. Include units. [2 points]

(b) Write the tangent-line model at $t = 2$ and use it to estimate $P(2.06)$. [3 points]

(c) Compare the one-sided secant estimates from 1.9 to 2 and from 2 to 2.1. What does the comparison suggest about $P''(2)$? [2 points]

(d) Explain why the table alone does not prove differentiability at $t = 2$. [1 points]

FRQ2 – Derivative values from a table

At $x = 1$, $f = 3$, $f' = -2$, $f'' = 5$, $g = 4$, $g' = 6$, and $g'' = -1$.

(a) Find $(fg)'(1)$ and $(f/g)'(1)$. [4 points]

(b) Find $(fg)''(1)$. [3 points]

(c) Write the tangent line to $h(x) = f(x)g(x)$ at $x = 1$.

[2 points]

FRQ3 – Derivative from first principles

Let $f(x) = \sqrt{x + 5}$.

(a) Use the limit definition, not a memorized rule, to find $f'(a)$ for $a > -5$.

[4 points]

(b) Find the tangent line at $x = 4$.

[2 points]

(c) State the domain of the derivative function and explain the excluded endpoint.

[2 points]

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